Solution:

Since the number of years is even(eight), we can equally divide the given data it two equal parts and obtain the averages of first four years and last four years.

Year	Production	Average
1990	15	$\frac{15+11+20+10}{4}=14$
1991	11	
1992	20	
1993	10	
1994	15	$\frac{15+25+35+30}{4}=26.25$
1995	25	
1996	35	
1997	30	

Table 9.2

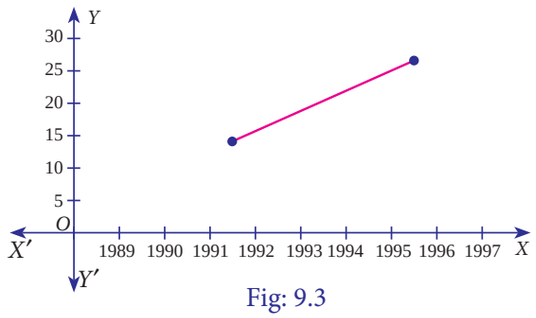


Fig: 9.3

Note

(i) The future values can be predicted.

(ii) The trend values obtained by this method and the predicted values are not precise.

9.1.3 Method of Moving Averages

Moving Averages Method gives a trend with a fair degree of accuracy. In this method, we take arithmetic mean of the values for a certain time span. The time span can be three- years, four -years, five- years and so on depending on the data set and our interest. We will see the working procedure of this method.

Procedure:

(i) Decide the period of moving averages (three- years, four -years).

(ii) In case of odd years, averages can be obtained by calculating,

$$\frac{a+b+c}{3}, \frac{b+c+d}{3}, \frac{c+d+e}{3}, \frac{d+e+f}{3}, \ldots$$

(iii) If the moving average is an odd number, there is no problem of centering it, the average value will be centered besides the second year for every three years.

(iv) In case of even years, averages can be obtained by calculating,

$$\frac{a+b+c+d}{4}, \frac{b+c+d+e}{4},$$

$$\frac{c+d+e+f}{4}, \frac{d+e+f+g}{4}, \ldots$$

(v) If the moving average is an even number, the average of first four values will be placed between 2nd and 3rd year, similarly the average of the second four values will be placed between 3rd and 4th year. These two averages will be again averaged and placed in the 3rd year. This continues for rest of the values in the problem. This process is called as centering of the averages.

Example 9.4

Calculate three-yearly moving averages of number of students studying in a higher secondary school in a particular village from the following data.

Year	Number of students
1995	332
1996	317
1997	357
1998	392
1999	402
2000	405
2001	410
2002	427
2003	435
2004	438

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